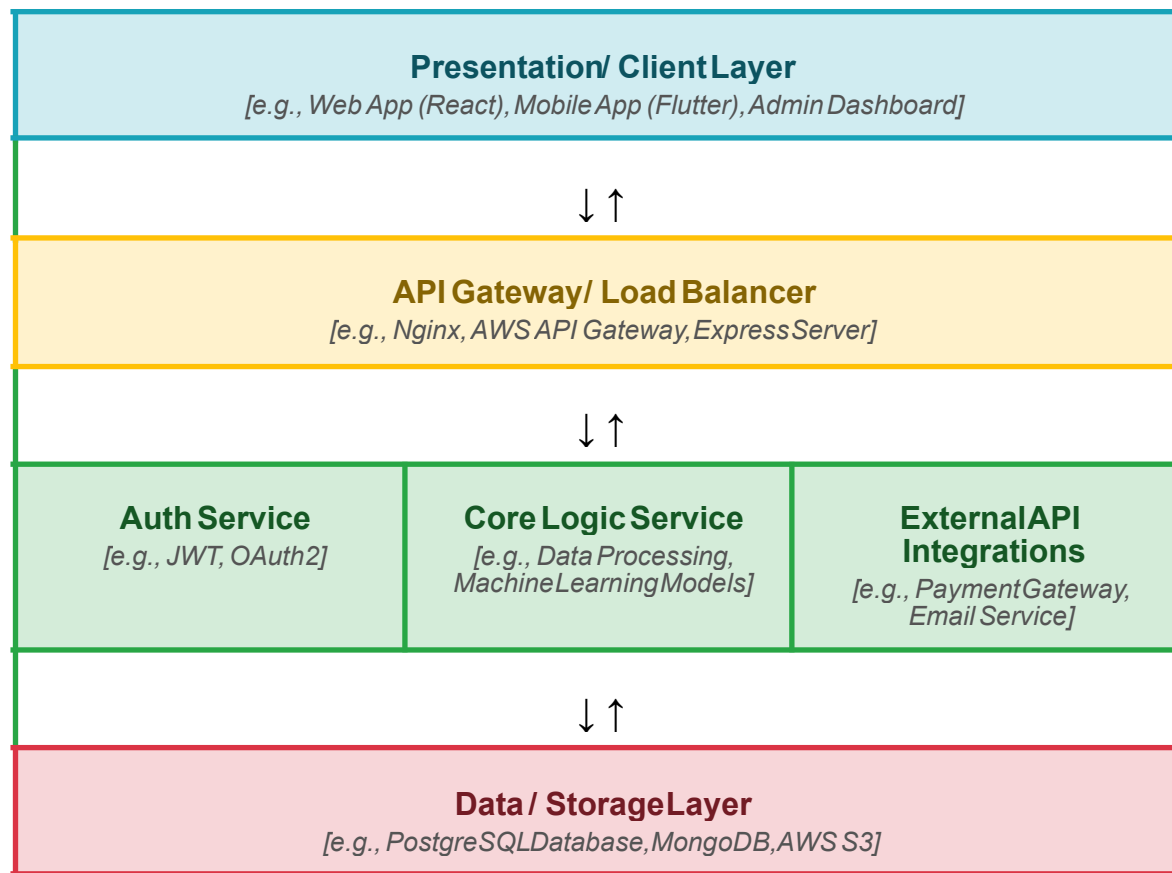


Solution Architecture

Date	30 June 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	This is the actual website screen that the borrower sees and clicks on. It displays the clean debt stress charts, has forms where users type in their loan balances or income numbers, and shows the final AI letters.	React.js
API Gateway	This acts as the secure traffic controller. It takes the loan numbers or letter requests typed by the user on the frontend screen and sends them instantly to the backend engine to be processed.	FastAPI Router / Uvicorn
Microservices / Core Services	Auth Service: Makes sure users can safely log into their accounts without anyone else stealing their private data. Core Logic Service: The main "brain" of the app. It takes the user's monthly income and bills, subtracts them to find the monthly surplus cash, and evaluates how severe their debt stress is.	Python / FastAPI Security / Internal Logic Modules
Database	The safe digital filing cabinet where the app permanently remembers each borrower's profile details, the specific banks they owe money to, and a complete history log of past generated settlement letters.	SQLite Database Engine with SQLAlchemy ORM